

Decision Trees.

1. What are the advantages and disadvantages of Decision Trees?

Ans: The main advantages of decision trees are that they are simple to understand, can handle both numbers and categories, and do not need scaling. The disadvantages are that they overfit easily, can be unstable with small changes in data, and are less accurate compared to ensembles.

2. What is entropy and how is it used in Decision Trees?

Ans: Entropy measures the impurity or disorder in a dataset. In decision trees, it is used to decide the best feature to split on by checking which split reduces entropy the most.

3. How does Information Gain work in Decision Trees?

Ans: Information Gain measures how much uncertainty is reduced after a split. The tree chooses the feature that provides the highest gain, meaning it separates the classes more effectively.

4. What is the stopping criterion for growing a Decision Tree?

Ans: A decision tree stops growing when no further useful splits can be made, when a maximum depth is reached, when a node becomes pure, or when too few samples remain. These criteria prevent unnecessary splits.

5. Why do Decision Trees tend to overfit?

Ans: Decision trees tend to overfit because they keep splitting until they fit every detail in the training data. This makes them capture noise instead of patterns, which reduces performance on unseen data.

6. How does a Decision Tree handle missing values?

Ans: Some decision tree algorithms handle missing values by finding alternative splits or by sending the sample down multiple branches with weights. Others may require preprocessing to fill or drop missing values.

7. Can Decision Trees handle both numerical and categorical data?

Ans: Yes, decision trees can handle both numerical and categorical data. They split numbers by thresholds and categories by grouping values into subsets.

8. How do Decision Trees handle imbalanced datasets?

Ans: Decision trees often favor the majority class in imbalanced datasets. To fix this, class weights, resampling, or balanced splitting criteria are used.

9. Why might a Decision Tree trained on noisy data give poor generalization?

Ans: If the training data contains noise, the tree may learn from those random errors and try to perfectly separate them. This leads to poor generalization and low accuracy on new data.

10. Can Decision Trees approximate nonlinear decision boundaries? How?

Ans: Decision trees can approximate nonlinear decision boundaries by recursively partitioning the feature space, trees create axis-aligned splits that approximate nonlinear boundaries.. With enough splits (depth), the resulting boundaries can mimic very complex shapes.

11. How would you interpret feature importance from a Decision Tree?

Ans: Feature importance in a decision tree is determined by how much a feature helps in reducing impurity across the tree. Features that are chosen near the top of the tree usually carry higher importance.

12. Calculation IG of a feature without any dataset provided, but with probability distribution.

Decision Tree

<u>X₁</u>	<u>X₂</u>	<u>Y</u>
3	1	0
2	1	1
3	1	1

Calculate IG of X₁.

Steps: Dataset entropy = $-\frac{1}{3}\log_2 \frac{1}{3} - \frac{2}{3}\log_2 \frac{2}{3}$
 $= 0.91$

Entropy of 3 = $-\frac{1}{2}\log_2 \frac{1}{2} - \frac{1}{2}\log_2 \frac{1}{2} = 1$

Entropy of 2 = $-\frac{1}{1}\log_2 \frac{1}{1} - \frac{0}{1}\log_2 \frac{0}{1}$
 $= 0$

$IG(X_1) = 0.91 - \frac{(2 \times 1)}{3} - \frac{1}{3} \times 0$
 $= 0.243$

Handwritten formulas on a piece of paper:

$$\text{Dataset entropy} = -P(0)\log_2(P(0)) - P(1)\log_2(P(1))$$
$$\text{Entropy of } Z = -P(0|3)\log_2(P(0|3)) - P(1|3)\log_2(P(1|3))$$

Naive Bayes

1: Why is it called “Naive”?

Ans: It is called “Naive” because the algorithm assumes that all features are independent of each other given the class label, which is usually not true in real-world data.

2: What are the main assumptions behind Naive Bayes?

Ans: The main assumptions are that features are conditionally independent given the class and that each feature contributes equally and independently to the final outcome.

3: What are the advantages and disadvantages of Naive Bayes?

Ans: The advantages are that it is simple, fast, works well on small datasets, and performs well for text classification. The disadvantages are that it relies on the independence assumption, struggles with highly correlated features, and may give poor estimates of probabilities.

4: Can Naive Bayes be used for both classification and regression?

Ans: Naive Bayes is mainly used for classification tasks. There are extensions for regression, but they are rarely used in practice compared to other regression models.

5: How does Naive Bayes handle continuous features?

Ans: Continuous features are usually handled using Gaussian Naive Bayes, which assumes the values follow a normal distribution within each class.

6: How does Naive Bayes handle zero probability issues?

Ans: It uses Laplace smoothing (also called additive smoothing), which adds a small value to all counts to avoid zero probabilities.

7: How does Naive Bayes handle missing values?

Ans: Some implementations skip the missing feature during probability calculation, while others use imputation methods or estimate probabilities based on available features.

8: Can Naive Bayes work with correlated features?

Ans: Naive Bayes can still run with correlated features, but the independence assumption gets violated, which may reduce accuracy.

9: How does Naive Bayes compare with logistic regression?

Ans: Naive Bayes is a generative model that learns class-conditional probabilities, while logistic regression is a discriminative model that learns decision boundaries. Logistic regression often performs better with correlated features, but Naive Bayes can be faster and works well with small datasets.

10: Why do Naive Bayes sometimes perform well even when independence assumption is violated?

Ans: It still produces good class predictions because the independence assumption mainly affects probability estimates, not the relative ranking of class probabilities.

Naive Bayes

Assume, we compute

$$\begin{aligned} P(0 | X_1 = x_1, X_2 = x_2, \dots) &= 0.3 \\ P(1 | X_1 = x_1, X_2 = x_2, \dots) &= 0.4 \end{aligned} \quad \left. \begin{array}{l} \text{Case 1:} \\ \text{conditional} \\ \text{independence} \end{array} \right\}$$

$$\begin{aligned} P(0 | X_1 = x_1, X_2 = x_2) &= 0.3 + e \\ P(1 | X_1 = x_1, X_2 = x_2) &= 0.4 + e \end{aligned} \quad \left. \begin{array}{l} \text{Case 2: No cond.} \\ \text{independence.} \end{array} \right\}$$

So, we can still distinguish classes.

11: Can Naive Bayes be used for multi-class classification?

Ans: Yes, Naive Bayes naturally supports multi-class problems by computing probabilities for each class and selecting the one with the highest value.

Logistic Regression

1: What is logistic regression and how is it different from Linear Regression?

Ans: Logistic regression is a statistical model used for classification problems where the output is categorical. Unlike linear regression, which predicts continuous values, logistic regression predicts probabilities that are mapped to classes using the sigmoid function.

2: What kind of problems can logistic regression solve?

Ans: Logistic regression is mainly used for binary classification problems such as spam detection, disease diagnosis, or churn prediction. It can also be extended to multi-class classification using multinomial logistic regression.

3: Why do we use the sigmoid function in logistic regression?

Ans: The sigmoid function maps any real-valued input into a range between 0 and 1, which can be interpreted as a probability. This makes it suitable for classification tasks where outputs must represent class probabilities.

4: What does the output of logistic regression represent?

Ans: The output represents the probability of a data point belonging to a certain class. Based on a threshold, usually 0.5, the probability is converted into a class prediction.

5: What are the assumptions behind logistic regression?

Ans: Logistic regression assumes that the relationship between features and the log-odds of the outcome is linear, that observations are independent, and that there is little or no multicollinearity among features.

6: What is the decision boundary in logistic regression?

Ans: The decision boundary is the surface that separates classes based on predicted probabilities. It is determined by the linear equation of features where the predicted probability equals the threshold, often 0.5.

7: What is the cost function used in logistic regression?

Ans: Logistic regression uses the log loss or cross-entropy cost function. This function measures the difference between the predicted probabilities and the actual class labels.

8: Why can't we use Mean Squared Error as a cost function in logistic regression?

Ans: Mean Squared Error is not suitable because the sigmoid function makes the relationship between input and output non-linear, causing optimization to become inefficient and unstable. Cross-entropy works better as it is convex and leads to faster, more reliable convergence.

Gradient Descent

1: How does Gradient Descent work?

Ans: Gradient Descent works by starting with an initial guess for the model parameters and then iteratively updating them in the opposite direction of the gradient of the cost function. With each step, the parameters move closer to the values that minimize the cost.

2: What is the role of the learning rate in Gradient Descent?

Ans: The learning rate controls the size of the steps taken toward the minimum. A good learning rate helps the algorithm converge faster without overshooting the minimum.

3: What happens if the learning rate is too high or too low, or negative?

Ans: If the learning rate is too high, the algorithm may overshoot and fail to converge. If it is too low, convergence will be very slow. A negative learning rate would move the parameters in the wrong direction, causing divergence.

4: How do you know if Gradient Descent is converging?

Ans: Convergence can be observed when the cost function values keep decreasing and eventually stabilize. Plotting the cost over iterations or monitoring when changes become very small helps confirm convergence.

5: Why is Gradient Descent preferred over solving equations directly?

Ans: For simple problems, equations can be solved directly, but in high-dimensional or complex models, direct solutions are computationally expensive or infeasible. Gradient Descent provides an efficient way to approximate the solution iteratively.

6: What is the difference between convex and non-convex functions in the context of Gradient Descent?

Ans: In convex functions, Gradient Descent always converges to the global minimum because the cost surface has a single bowl-shaped minimum. In non-convex functions, there may be many local minima or saddle points, so Gradient Descent might not reach the global minimum.

7: What is the cost function, and how does Gradient Descent minimize it?

Ans: The cost function measures how far the model's predictions are from the actual values. Gradient Descent minimizes it by calculating the gradient and updating parameters step by step to reduce the error.

8: What are common problems faced in Gradient Descent optimization?

Ans: Common problems include slow convergence, getting stuck in local minima or saddle points, vanishing or exploding gradients, and sensitivity to the choice of learning rate.

9: Does Gradient Descent always find the global minima?

Ans: No, it only guarantees reaching the global minimum for convex functions. In non-convex

problems like deep neural networks, it may converge to local minima or saddle points, though these solutions are often good enough in practice.

10. Assume a function $f(x) = 3x^2 + 5x$. Assume, the initial value of x is 0.5, and learning rate is 0.001. Perform one iteration of gradient descent to minimize the function (find updated value of x).

Ans: Given, initial value for $x = 0.5$

$$f'(x) = 6x + 5$$

$$f'(0.5) = 6 * (0.5) + 5 = 8$$

$$x' = x - (f'(x) * \alpha) \text{ [Parameter updation formula for gradient descent]}$$

$$x' = 0.5 - (8 * 0.001) = 0.492 \text{ (Ans)}$$

Test

Look at the function value at point $x = 0.5$, and $x = 0.492$. You will see that $f(0.5) > f(0.492)$.

$$f(0.5) = 3(0.5)^2 + 5(0.5) = 3.25$$

$$f(0.492) = 3(0.492)^2 + 5(0.492) = 3.18$$

Take sequential preparation.

1. Decision Tree, Naive Bayes
2. Gradient descent, Perceptron, Logistic Regression, Learning Rate..
3. Gradient descent simply explained (from 28 minutes).
4. Probability.